

CAMBRIDGE INTERNATIONAL AS & A LEVEL CHEMISTRY (9701)

Topic 12 — Nitrogen and Sulfur

Topical Past Paper Question Pack & Comprehensive Mark Scheme

CANDIDATE	Urwah
SYLLABUS	Cambridge International AS Level Chemistry (9701)
TOPIC COVERAGE	12.1 Nitrogen & Ammonia · 12.2 Nitrogen Oxides in Atmosphere · 12.3 Sulfur & The Contact Process · Acid Rain
EXAM SERIES	Past Paper Compilation (2019 – 2025 Paper 2 Theory & Paper 1)
TOTAL MARKS	173 Marks (50 Questions)

HOW TO USE THIS PACK

1. Complete each question carefully in the answer spaces provided, paying close attention to mark allocations.
2. Questions are classified into Easy and Hard tiers to guide progression from core recall to advanced data analysis.
3. Every question includes authentic Cambridge examination codes (e.g. 9701/22/M/J/22) for cross-referencing.
4. Full examiner marking points with point-by-point breakdowns are provided at the end of the document.

SYLLABUS SUBTOPICS COVERED

12.1 Nitrogen, Ammonia & Fertilisers— Unreactivity of N₂ (triple bond), basicity of NH₃, ammonium ion structure (dative bonding), fertilisers**12.2 Nitrogen Oxides & Smog**

— Formation of NO in car engines, catalytic converters (Pt/Rh/Pd honeycomb), tropospheric ozone depletion

12.3 Sulfur & The Contact Process— Manufacture of H₂SO₄ by the Contact process (V₂O₅ catalyst, 450 °C, 1–2 atm, oleum), acid rain

SUBTOPIC 12.1 — NITROGEN, AMMONIA, FERTILISERS & EUTROPHICATION

1 Unreactivity of Nitrogen and Structure of Ammonia — 9701/21/M/J/23/Q3(a)–(c)

EASY · Syllabus 12.1

Nitrogen gas constitutes approximately 78% of the Earth's atmosphere, yet it is chemically very unreactive under ambient conditions.

((a)) Explain why nitrogen gas is remarkably unreactive at room temperature, referring to its bonding and bond enthalpy. [2]

.....

.....

.....

((b)) Draw a dot-and-cross diagram of the ammonia molecule, NH_3 . State its molecular shape and the H-N-H bond angle. [3]

.....

.....

.....

.....

((c)) Explain why ammonia acts as a Brønsted–Lowry base in aqueous solution. [2]

.....

.....

.....

2 Formation and Structure of the Ammonium Ion — 9701/22/O/N/22/Q4(a)–(c)

EASY · Syllabus 12.1

When ammonia gas dissolves in hydrochloric acid, the ammonium ion, NH_4^+ , is formed.

((a)) Explain the type of bond formed between the nitrogen atom of an ammonia molecule and a hydrogen ion (H^+). [2]

.....

.....

.....

((b)) State the shape and bond angle of the ammonium ion, NH_4^+ , and explain why the bond angle differs from that in ammonia. [3]

.....

.....

.....

.....

((c)) Write an ionic equation for the reaction between aqueous ammonium sulfate and aqueous sodium hydroxide, and state the observation when the mixture is warmed. [2]

.....

.....

.....

3 Displacement of Ammonia from Ammonium Salts — 9701/23/M/J/22/Q3(a)–(b)

EASY · Syllabus 12.1

Ammonium salts react with strong bases upon warming to release ammonia gas.

((a)) A sample of solid ammonium chloride is heated with solid calcium hydroxide. Write the balanced chemical equation for this reaction. [2]

.....

.....

.....

- ((b)) Explain why slaked lime, $\text{Ca}(\text{OH})_2$, should not be applied to agricultural land at the same time as ammonium nitrate fertiliser. [2]

.....

.....

.....

4 Percentage Nitrogen in Fertilisers — 9701/21/M/J/21/Q2(b)

EASY · Syllabus 12.1

Ammonium nitrate, NH_4NO_3 , and urea, $\text{CO}(\text{NH}_2)_2$, are two common synthetic nitrogenous fertilisers.

- ((a)) Calculate the percentage by mass of nitrogen in pure ammonium nitrate. (Ar: H = 1.0, N = 14.0, O = 16.0) [2]

.....

.....

.....

- ((b)) Calculate the percentage by mass of nitrogen in pure urea, $\text{CO}(\text{NH}_2)_2$. (Ar: C = 12.0, H = 1.0, N = 14.0, O = 16.0) [2]

.....

.....

.....

- ((c)) Suggest one practical advantage of using urea over ammonium nitrate as a solid commercial fertiliser. [1]

.....

.....

5 Environmental Eutrophication Caused by Fertilisers — 9701/22/M/J/21/Q3(c)-(e)

EASY · Syllabus 12.1

Excessive application of artificial nitrogenous fertilisers to agricultural land can lead to serious environmental damage in nearby rivers and lakes.

- ((a)) Explain why nitrate fertilisers are more prone to leaching from soil into watercourses than phosphate or potassium fertilisers. [1]

.....

.....

- ((b)) Outline the four chronological stages of eutrophication that occur when leached nitrates enter a freshwater lake. [4]

.....

.....

.....

.....

- ((c)) Explain why fish and other aquatic animals die as a result of eutrophication. [1]

.....

.....

6 Haber Process Conditions and Economics — 9701/11/M/J/22/Q20

EASY · Syllabus 12.1

Ammonia is manufactured by the Haber process: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ $\Delta H = -92 \text{ kJ mol}^{-1}$.

0

- Which set of operating conditions represents the standard industrial compromise?
- A. 100 °C, 1 atm, no catalyst
 - B. 450 °C, 200 atm, iron catalyst
 - C. 800 °C, 500 atm, platinum catalyst
 - D. 25 °C, 1000 atm, nickel catalyst

[1]

7

Industrial Manufacture of Nitric Acid (Ostwald Process) — 9701/22/O/N/21/Q4(a)–(c)

HARD · Syllabus 12.1

Ammonia is oxidised in the Ostwald process to produce nitric acid. Stage 1 involves the catalytic oxidation of ammonia: $4\text{NH}_3(\text{g}) + 5\text{O}_2(\text{g}) \rightarrow 4\text{NO}(\text{g}) + 6\text{H}_2\text{O}(\text{g})$.

((a))

State the catalyst and typical temperature used in Stage 1 of the Ostwald process.

[2]

((b))

State the oxidation number of nitrogen in NH_3 , NO , and HNO_3 .

[3]

((c))

Write a balanced chemical equation for the reaction of nitrogen dioxide with water and oxygen in the final stage of nitric acid manufacture.

[1]

8

Bonding in Ammonium Nitrate — 9701/12/F/M/22/Q18

EASY · Syllabus 12.1

Which types of bonding are present in solid ammonium nitrate, NH_4NO_3 ?

0

- Select the correct combination:
- A. Ionic bonding only
 - B. Covalent bonding only
 - C. Ionic, covalent, and dative covalent bonding
 - D. Covalent and metallic bonding

[1]

9

Thermal Decomposition of Ammonium Salts — 9701/21/O/N/20/Q3(a)–(b)

HARD · Syllabus 12.1

Different ammonium salts undergo distinct decomposition reactions when heated strongly.

((a))

When ammonium chloride, NH_4Cl , is heated, it undergoes reversible thermal dissociation. Write the equation and state the observations.

[2]

- ((b))** When ammonium nitrate, NH_4NO_3 , is heated carefully, it decomposes into dinitrogen monoxide (nitrous oxide) and steam. Write the balanced equation and deduce the oxidation state change for each nitrogen atom. **[3]**

.....

.....

.....

.....

10 Qualitative Test for the Ammonium Ion — 9701/23/O/N/21/Q2(a)-(b)

EASY · Syllabus 12.1

A student is given an unlabelled white solid suspected to be an ammonium salt.

- ((a))** Describe a chemical test to confirm the presence of ammonium ions in the sample. **[2]**

.....

.....

.....

- ((b))** Explain why damp litmus paper must be used rather than dry litmus paper. **[1]**

.....

.....

11 Ammonia as a Ligand in Complex Formation — 9701/22/M/J/20/Q3(a)-(b)

HARD · Syllabus 12.1

When aqueous ammonia is added dropwise until in excess to a solution of copper(II) sulfate, a series of colour changes occurs.

- ((a))** State the observation when a small amount of aqueous ammonia is added, and identify the precipitate. **[2]**

.....

.....

.....

- ((b))** State the observation when excess aqueous ammonia is added, and give the formula of the complex ion formed. **[2]**

.....

.....

.....

12 Reactivity Comparison: N_2 vs CO — 9701/13/M/J/22/Q19

HARD · Syllabus 12.1

Both nitrogen, N_2 , and carbon monoxide, CO , are isoelectronic with 14 electrons and contain a triple bond between two atoms.

- 0** Why is carbon monoxide significantly more reactive than nitrogen towards transition metals? **[1]**
- A. CO has a much lower triple bond enthalpy than N_2 .
 - B. CO has a permanent dipole and an accessible lone pair on carbon that acts as a strong sigma-donor and pi-acceptor ligand.
 - C. CO is an ionic compound.
 - D. N_2 has no lone pairs of electrons.

.....

13 Titration Analysis of Ammonium Sulfate Fertiliser — 9701/21/M/J/22/Q3(d)

HARD · Syllabus 12.1

A 1.500 g sample of impure ammonium sulfate fertiliser was boiled with 50.0 cm^3 of $1.000 \text{ mol dm}^{-3} \text{ NaOH(aq)}$ until all ammonia gas was expelled. The remaining unreacted NaOH required 26.80 cm^3 of $0.500 \text{ mol dm}^{-3} \text{ HCl(aq)}$ for neutralisation. (M_r of $(\text{NH}_4)_2\text{SO}_4 = 132.1$)

- ((a)) Calculate the initial number of moles of NaOH added. [1]
-
-
- ((b)) Calculate the number of moles of unreacted excess NaOH. [1]
-
-
- ((c)) Calculate the percentage purity of ammonium sulfate in the fertiliser sample. [3]
-
-
-
-

SUBTOPIC 12.2 — NITROGEN OXIDES IN THE ATMOSPHERE & CATALYTIC CONVERTERS

14 Three-Way Catalytic Converters in Motor Vehicles — 9701/22/M/J/23/Q4(a)–(c)

HARD · Syllabus 12.2

Modern petrol-powered cars are fitted with catalytic converters to minimise harmful exhaust emissions. Fig. 14.1 illustrates the honeycomb converter and reactions.

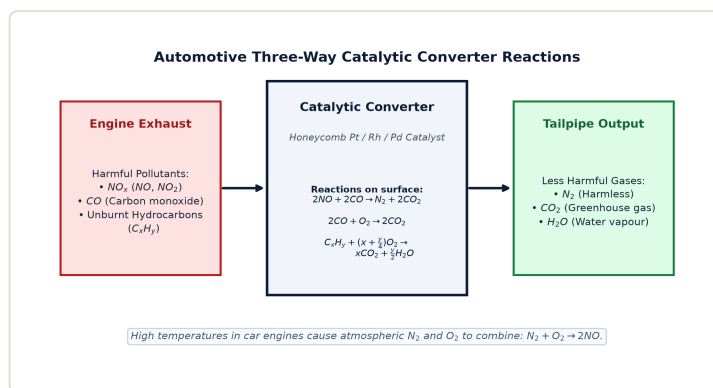


Fig. 14.1: Reactions occurring across a three-way automotive catalytic converter.

- ((a)) Explain how nitrogen monoxide, NO , is formed in an internal combustion engine, and write a balanced equation. [2]
-
-
-
- ((b)) Write the balanced chemical equation for the simultaneous removal of nitrogen monoxide and carbon monoxide inside a catalytic converter. [1]
-
-
-
- ((c)) Name two transition metals used as catalysts in the converter, and explain why the catalyst is deposited as a thin washcoat on a ceramic honeycomb structure. [3]
-
-
-
-

15 Atmospheric Photochemical Smog Formation — 9701/21/O/N/22/Q4(a)–(c)

HARD · Syllabus 12.2

In urban areas with dense traffic, nitrogen dioxide contributes to the generation of photochemical smog on sunny days.

((a)) State the color of nitrogen dioxide gas, NO₂. **[1]**

.....

.....

((b)) Write equations showing how NO₂ absorbs ultraviolet sunlight to produce reactive oxygen radicals, and how these radicals react with molecular oxygen to form tropospheric ozone, O₃. **[2]**

.....

.....

.....

((c)) State two adverse environmental or human health effects associated with tropospheric ozone and photochemical smog. **[2]**

.....

.....

.....

16 Formation of Nitric Acid in Rainwater — 9701/11/M/J/23/Q20

EASY · Syllabus 12.2

Which sequence of reactions represents the natural formation of acid rain caused by lightning?

() **[1]**

Select the correct sequence:

- A. N₂ + O₂ → 2NO; 2NO + O₂ → 2NO₂; 4NO₂ + 2H₂O + O₂ → 4HNO₃
- B. N₂ + 2O₂ → 2NO₂; NO₂ + H₂O → HNO₂ + H₂
- C. N₂ + 3H₂ → 2NH₃; 2NH₃ + 4O₂ → 2HNO₃ + 2H₂O
- D. 2NO + H₂O → HNO₂ + HNO₃

.....

17 Catalytic Converter Poisoning by Lead — 9701/22/F/M/21/Q4(c)

EASY · Syllabus 12.2

Cars fitted with catalytic converters must strictly use unleaded petrol.

() Explain why leaded petrol causes permanent loss of activity in a catalytic converter. **[2]**

.....

.....

.....

18 Dimerisation Equilibrium of Nitrogen Dioxide — 9701/23/M/J/21/Q4(a)–(c)

HARD · Syllabus 12.2

Brown nitrogen dioxide gas exists in dynamic equilibrium with colorless dinitrogen tetroxide: 2NO₂(g) ⇌ N₂O₄(g) ΔH = -57 kJ mol⁻¹.

((a)) A gas syringe containing an equilibrium mixture of NO₂ and N₂O₄ is plunged into a beaker of hot water. State the observed color change and explain it using Le Chatelier's principle. **[2]**

.....

.....

.....

- ((b))** The plunger of the syringe is suddenly pushed in, halving the gas volume. Describe the immediate color change and the subsequent color change as new equilibrium is established. **[3]**

.....

.....

.....

.....

19 Free Radical Mechanism of Ozone Formation by NO₂ — 9701/21/M/J/20/Q4(d)

HARD · Syllabus 12.2

In the troposphere, NO₂ acts as a primary photochemical precursor to ground-level ozone.

- ((a))** Explain the meaning of the term free radical. **[1]**

.....

.....

- ((b))** Explain why ozone is beneficial in the stratosphere but considered a harmful pollutant in the troposphere. **[2]**

.....

.....

.....

20 Combustion Reactions in Jet Engines — 9701/12/O/N/22/Q19

EASY · Syllabus 12.2

Aviation jet engines operating at high altitudes produce significant quantities of nitrogen oxides.

- 0** Which factor is primarily responsible for the synthesis of nitrogen oxides in jet aircraft exhausts? **[1]**
- A. Nitrogen-containing additives in aviation kerosene.
 - B. The high operating temperature in the combustion chamber causing atmospheric N₂ and O₂ to combine.
 - C. Low atmospheric pressure at high altitude.
 - D. Catalytic effect of the turbine blades.

.....

21 Role of NO in Catalytic Destruction of Stratospheric Ozone — 9701/22/O/N/23/Q4(a)–(c)

HARD · Syllabus 12.2

Nitrogen monoxide can act as a homogeneous catalyst for the breakdown of ozone in the stratosphere.

- ((a))** Write the two elementary propagation steps showing how NO catalyses the conversion of O₃ and an oxygen radical into O₂. **[2]**

.....

.....

.....

- ((b))** Construct the overall equation for this catalytic cycle. **[1]**

.....

.....

- ((c))** Explain why NO is classified as a homogeneous catalyst in this process. **[1]**

.....

.....

22 Exhaust Gas Composition Analysis — 9701/21/M/J/23/Q4(e)

HARD · Syllabus 12.2

A 100 cm³ sample of exhaust gas from a poorly tuned engine contains 4.0 cm³ of carbon monoxide and 2.0 cm³ of nitrogen monoxide. The gas is passed over a heated platinum catalyst.

- ((a))** Assuming complete reaction according to $2\text{NO}(\text{g}) + 2\text{CO}(\text{g}) \rightarrow \text{N}_2(\text{g}) + 2\text{CO}_2(\text{g})$, calculate the volume of CO remaining unreacted. **[2]**

.....

.....

.....

- ((b))** Calculate the volume of N₂ gas produced by this reaction. **[1]**

.....

.....

23 Environmental Regulations on Vehicle Emissions — 9701/13/O/N/21/Q19

EASY · Syllabus 12.2

Which emission from petrol vehicle tailpipes is NOT reduced by a catalytic converter?

- ()** Select the gas: **[1]**
- A. Carbon monoxide
 - B. Nitrogen monoxide
 - C. Carbon dioxide
 - D. Unburnt octane

.....

24 Formation of PAN in Photochemical Smog — 9701/22/F/M/22/Q3(d)

HARD · Syllabus 12.2

Peroxyacetyl nitrate (PAN), CH₃CO-OO-NO₂, is a secondary pollutant formed in photochemical smog by the reaction of hydrocarbons, oxygen, and nitrogen dioxide.

- ((a))** Distinguish between a primary pollutant and a secondary pollutant. **[2]**

.....

.....

.....

- ((b))** State the main physiological hazard posed by PAN to humans. **[1]**

.....

.....

25 Acid Rain Formation: Homogeneous vs Heterogeneous Catalysis — 9701/23/O/N/22/Q3(c)

HARD · Syllabus 12.2

The atmospheric conversion of SO₂ to SO₃ is catalysed by NO₂ in the gas phase and by soot particulates.

- ((a))** Classify the catalysis by NO₂ as homogeneous or heterogeneous and justify your choice. **[2]**

.....

.....

.....

- ((b))** Classify the catalysis by airborne soot particles and describe the first physical step of this mechanism. **[2]**

.....

.....

.....

26 MCQ on Nitrogen Oxides Sources — 9701/11/F/M/23/Q18

EASY · Syllabus 12.2

Which natural event contributes significantly to the formation of atmospheric nitrogen monoxide?

0

Select the correct natural event:

- A. Volcanic eruptions releasing hydrogen sulfide
- B. Lightning discharges providing high activation energy
- C. Evaporation of ocean spray
- D. Respiration by aerobic microorganisms

[1]

SUBTOPIC 12.3 — SULFUR, THE CONTACT PROCESS & ACID RAIN

27 The Contact Process: Equilibrium and Compromise Conditions — 9701/21/M/J/23/Q2(a)-(d)

HARD · Syllabus 12.3

Sulfuric acid is manufactured on an enormous industrial scale by the Contact Process. Fig. 27.1 displays the three key stages.

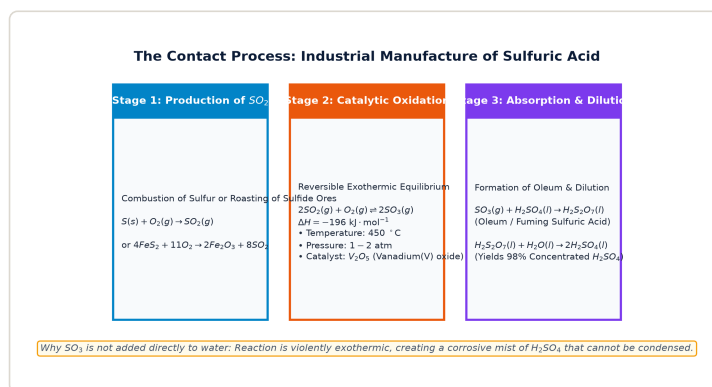


Fig. 27.1: Flowchart and operating conditions for the Contact Process.

((a))

Write the balanced reversible equation for Stage 2 of the Contact Process, including state symbols and the enthalpy change delta-H.

[2]

((b))

Explain why an operating temperature of approximately 450 °C is chosen, rather than a much lower temperature such as 100 °C.

[2]

((c))

Explain why the process is operated at near-atmospheric pressure (1–2 atm), even though Le Chatelier's principle indicates higher pressure increases the yield of SO₃.

[2]

((d))

State the chemical formula of the catalyst used in Stage 2 and explain how it affects the equilibrium yield of SO₃.

[2]

28 Absorption Stage of the Contact Process (Oleum) — 9701/22/M/J/22/Q3(a)–(c)

HARD · Syllabus 12.3

In Stage 3 of the Contact Process, sulfur trioxide is absorbed into 98% concentrated sulfuric acid.

((a)) Explain why sulfur trioxide is NOT absorbed directly into liquid water. **[2]**

.....

.....

.....

((b)) Write the balanced equation for the reaction of sulfur trioxide with concentrated sulfuric acid to form oleum. **[1]**

.....

.....

((c)) Write the balanced equation for the dilution of oleum with water to produce concentrated sulfuric acid. **[1]**

.....

.....

29 Atmospheric Catalysis of SO₂ by NO₂ — 9701/21/O/N/21/Q3(a)–(c)

HARD · Syllabus 12.3

Sulfur dioxide emitted into the atmosphere is oxidised to sulfur trioxide, catalysed by atmospheric nitrogen dioxide.

((a)) Write two equations representing the catalytic cycle by which NO₂ catalyses the oxidation of SO₂ to SO₃. **[2]**

.....

.....

.....

((b)) Construct the overall equation for this oxidation. **[1]**

.....

.....

((c)) Write the equation for the formation of sulfuric acid when sulfur trioxide dissolves in atmospheric raindrops. **[1]**

.....

.....

30 Environmental Damage from Acid Rain — 9701/23/M/J/21/Q2(c)–(e)

EASY · Syllabus 12.3

Rainwater with a pH below 5.6 is classified as acid rain, primarily caused by sulfuric and nitric acids.

((a)) Write a chemical equation showing how acid rain corrodes stonework made of limestone or marble. **[1]**

.....

.....

((b)) Explain how acid rain harms freshwater fish populations in natural lakes. **[2]**

.....

.....

.....

((c)) Describe how acid rain damages forest trees and soil fertility. **[2]**

.....

.....

.....

31 Flue-Gas Desulfurisation (FGD) by Calcium Carbonate — 9701/22/O/N/20/Q4(a)–(c)

HARD · Syllabus 12.3

Coal-fired electricity power stations use flue-gas desulfurisation units to remove sulfur dioxide before exhaust gases leave the smokestack.

((a)) Write a balanced chemical equation for the reaction between calcium carbonate and sulfur dioxide. **[1]**

.....

.....

((b)) The calcium sulfite formed is reacted with air and water to produce calcium sulfate dihydrate (gypsum), $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$. Write the balanced equation. **[2]**

.....

.....

.....

((c)) Calculate the mass of pure CaCO_3 required to react completely with 64.0 tonnes of SO_2 gas. (Mr: $\text{CaCO}_3 = 100.1$, $\text{SO}_2 = 64.1$) **[2]**

.....

.....

.....

32 Sulfur Dioxide as a Food Preservative — 9701/11/M/J/21/Q20

EASY · Syllabus 12.3

Sulfur dioxide, SO_2 , is widely used in winemaking and the preservation of dried fruits.

0 Which pair of chemical properties makes SO_2 particularly suitable as a food preservative? **[1]**

A. It acts as an antioxidant and as an antimicrobial agent.

B. It is a strong oxidising agent and sweetening agent.

C. It neutralises alkaline food components and speeds up fermentation.

D. It decomposes into harmless nitrogen gas.

.....

33 Chemical Test for Sulfur Dioxide Gas — 9701/22/F/M/20/Q2(b)

EASY · Syllabus 12.3

Sulfur dioxide is a toxic, colourless, choking gas.

((a)) Describe a chemical test to confirm the presence of sulfur dioxide gas, stating the reagent and the observed color change. **[2]**

.....

.....

.....

((b)) Write the ionic equation for the redox reaction occurring in this test. **[2]**

.....

.....

.....

34 Roasting of Zinc Blende to Produce SO_2 — 9701/23/O/N/21/Q3(d)

HARD · Syllabus 12.3

Zinc blende consists predominantly of zinc sulfide, ZnS . In metallurgy, zinc blende is roasted in air to produce zinc oxide and sulfur dioxide.

((a)) Write the balanced chemical equation for the roasting of zinc sulfide in excess air. [2]

.....

.....

.....

((b)) State how modern metallurgical smelters utilize the SO₂ gas produced to prevent environmental pollution. [1]

.....

.....

35 Oxidation States in the Contact Process — 9701/12/M/J/22/Q20

EASY · Syllabus 12.3

In the Contact Process, the conversion of sulfur to sulfuric acid involves several changes in oxidation state.

() Which row correctly gives the oxidation state of sulfur in each compound? [1]

- A. S: 0; SO₂: +4; SO₃: +6; H₂SO₄: +6
- B. S: 0; SO₂: +2; SO₃: +4; H₂SO₄: +7
- C. S: -2; SO₂: +4; SO₃: +6; H₂SO₄: +6
- D. S: 0; SO₂: +4; SO₃: +3; H₂SO₄: +6

.....

36 Acid-Base and Redox Properties of Concentrated H₂SO₄ — 9701/21/M/J/21/Q3(d)

HARD · Syllabus 12.3

Concentrated sulfuric acid can act as a strong acid, an oxidising agent, and a dehydrating agent.

((a)) When concentrated H₂SO₄ is added to hydrated copper(II) sulfate crystals, the blue crystals turn white. Name the property of sulfuric acid demonstrated by this change. [1]

.....

.....

((b)) When concentrated H₂SO₄ is added to solid sucrose, C₁₂H₂₂O₁₁, a black column of carbon rises with steam. Write the chemical equation for this dehydration. [1]

.....

.....

((c)) When copper metal is heated with concentrated H₂SO₄, effervescence occurs and a blue solution forms. Write the balanced equation and state which property of H₂SO₄ is demonstrated. [2]

.....

.....

.....

37 Deducing the Structure of Sulfur Dioxide — 9701/13/O/N/22/Q18

HARD · Syllabus 12.3

Consider the sulfur dioxide molecule, SO₂.

() What is the shape and approximate bond angle of the SO₂ molecule? [1]

- A. Linear; 180°
- B. Bent (non-linear); 119°
- C. Trigonal planar; 120°
- D. Pyramidal; 107°

.....

38 Volumetric Titration of SO₂ in Wine — 9701/22/M/J/23/Q3(e)

HARD · Syllabus 12.3

The free SO₂ in a 50.0 cm³ sample of white wine was determined by titration against 0.00500 mol dm⁻³ aqueous iodine, I₂, using starch indicator. The reaction is: $\text{SO}_2 + \text{I}_2 + 2\text{H}_2\text{O} \rightarrow \text{SO}_4^{2-} + 2\text{I}^- + 4\text{H}^+$. The titre required was 12.80 cm³.

((a)) State the color change observed at the end-point of this titration. **[1]**

.....

.....

((b)) Calculate the mass of free SO₂ present in 1.0 dm³ of this wine in milligrams (mg). (Mr of SO₂ = 64.1) **[3]**

.....

.....

.....

.....

39 Comparison of Oxides: CO₂ vs SO₂ — 9701/11/O/N/21/Q19

EASY · Syllabus 12.3

Both carbon dioxide and sulfur dioxide are acidic non-metal oxides.

(i) Which reagent can be used to distinguish between gaseous CO₂ and SO₂? **[1]**

- A. Calcium hydroxide solution (limewater)
- B. Acidified potassium manganate(VII) solution
- C. Universal indicator solution
- D. Sodium hydroxide solution

.....

40 Industrial Uses of Sulfuric Acid — 9701/21/M/J/20/Q2(c)

EASY · Syllabus 12.3

Sulfuric acid is frequently described as the 'king of chemicals' due to its ubiquitous industrial use.

((a)) State the single largest worldwide commercial use of sulfuric acid. **[1]**

.....

.....

((b)) Write the chemical equation for the manufacture of the nitrogenous fertiliser ammonium sulfate from sulfuric acid and ammonia. **[1]**

.....

.....

((c)) State two other distinct commercial applications of sulfuric acid. **[2]**

.....

.....

.....

41 Natural and Anthropogenic Sulfur Cycles — 9701/12/F/M/21/Q18

EASY · Syllabus 12.3

Which natural event contributes significant emissions of sulfur dioxide into the atmosphere?

0

- Select the major natural source:
- A. Volcanic eruptions
 - B. Evaporation from oceans
 - C. Respiration by forests
 - D. Agricultural crop cultivation

[1]

42

Enthalpy Profile of the Contact Process Reaction — 9701/23/M/J/22/Q2(d)

HARD · Syllabus 12.3

The catalytic oxidation of SO_2 to SO_3 has an activation energy of E_a (uncatalysed) = 250 kJ mol^{-1} and E_a (catalysed) = 80 kJ mol^{-1} . The overall reaction is exothermic ($\Delta H = -196 \text{ kJ mol}^{-1}$).

((a))

Sketch or describe the reaction pathway diagram showing curves for both the catalysed and uncatalysed reactions, clearly labelling E_a and ΔH .

[3]

((b))

Explain in terms of collision theory why V_2O_5 significantly increases the rate of reaction at 450°C .

[2]

43

Equilibrium Expression and K_c for Contact Process — 9701/22/O/N/21/Q2(e)

HARD · Syllabus 12.3

Consider the equilibrium: $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$.

((a))

Write the equilibrium expression for K_c and state its units.

[2]

((b))

At 700 K , a 2.0 dm^3 vessel contains 0.40 mol of SO_2 , 0.20 mol of O_2 , and 1.60 mol of SO_3 at equilibrium. Calculate the value of K_c .

[3]

44

Acid Rain pH and Hydronium Ion Concentration — 9701/12/M/J/23/Q20

EASY · Syllabus 12.3

Unpolluted rainwater naturally has a pH of approximately 5.6 due to dissolved carbon dioxide. In an industrial area affected by acid rain, the pH was measured as 3.6.

0

- By what factor has the concentration of hydrogen ions, $[\text{H}^+]$, increased in the acid rain sample compared to unpolluted rainwater?
- A. 2 times
 - B. 20 times
 - C. 100 times
 - D. 1000 times

[1]

45 Chemical Weathering of Marble Statues — 9701/21/O/N/23/Q3(c)

HARD · Syllabus 12.3

Historical statues sculpted from marble (CaCO_3) in European cities have suffered extensive erosion since the industrial revolution.

- ((a))** Explain why calcium sulfate formed on marble surfaces causes crumbling and spalling of stone rather than acting as a protective coating. **[2]**

.....

.....

.....

- ((b))** Write the ionic equation for the reaction of calcium carbonate with nitric acid from acid rain. **[1]**

.....

.....

46 Comparison of Acid Rain Pollutants: SO_2 vs NO_x — 9701/13/M/J/21/Q20

EASY · Syllabus 12.3

Which human activity is the predominant source of anthropogenic sulfur dioxide emissions worldwide?

- ()** Select the primary activity: **[1]**
- A. Petrol car transport
 - B. Combustion of coal and heavy fuel oil in thermal power stations
 - C. Manufacture of ammonia fertilisers
 - D. Domestic natural gas heating

.....

47 Balancing Atmospheric Redox Equations — 9701/22/F/M/23/Q3(d)

HARD · Syllabus 12.3

In clouds, sulfur dioxide can also be oxidised by hydrogen peroxide, H_2O_2 .

- ((a))** Write a balanced equation for the aqueous oxidation of SO_2 by H_2O_2 . **[1]**

.....

.....

- ((b))** Deduce the changes in oxidation number for sulfur and oxygen in this reaction. **[2]**

.....

.....

.....

48 Liming of Acidified Scandinavian Lakes — 9701/21/M/J/22/Q4(d)

EASY · Syllabus 12.3

In Sweden and Norway, helicopters have been used to spray thousands of tonnes of powdered limestone (CaCO_3) over acidified lakes.

- ((a))** Explain how the addition of limestone helps restore the biological ecosystem of an acidified lake. **[2]**

.....

.....

.....

- ((b))** Explain why liming is considered an emergency remediation measure rather than a long-term solution to acid rain. **[1]**

.....

.....

49 MCQ on Contact Process Catalyst — 9701/11/M/J/22/Q19

EASY · Syllabus 12.3

What is the catalyst used in the Contact process, and what is its oxidation state?

0

Select the correct catalyst and oxidation state:

- A. Iron; +3
- B. Vanadium(V) oxide; +5
- C. Platinum; 0
- D. Nickel; +2

[1]

50 Synoptic Problem: Industrial and Atmospheric Sulfur Chemistry — 9701/21/O/N/23/Q5(a)-(e)

HARD · Syllabus 12.3

A power station burns coal containing 1.60% sulfur by mass. The station burns 10,000 tonnes of coal per day. All sulfur is converted to SO_2 .

((a))

Calculate the mass of sulfur dioxide produced per day in tonnes. (Ar: S = 32.1, O = 16.0)

[2]

((b))

If 95% of this SO_2 is removed by a wet limestone FGD scrubber, calculate the mass of gypsum, $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$, produced per day. (Mr: $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ = 172.2)

[3]

((c))

The remaining 5% of SO_2 escapes into the atmosphere where it is catalysed by NO_2 into sulfuric acid rain. Write the two equations representing this catalytic cycle.

[2]

Mark Scheme — Topic 12 — Nitrogen and Sulfur

CAMBRIDGE INTERNATIONAL AS LEVEL CHEMISTRY (9701) — THEORY MARKING POINTS

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
1(a)	Nitrogen molecules contain a triple covalent bond (NN) with a very high bond enthalpy (945 kJ mol ⁻¹) [1] Significant energy is required to break the strong triple bond / the reaction has a very high activation energy [1]	2
1(b)	Dot-and-cross diagram showing three shared N-H pairs and one lone pair on N [1] Shape: Trigonal pyramidal [1] Bond angle: 107° [1]	3
1(c)	Ammonia has a non-bonding lone pair of electrons on the nitrogen atom [1] It accepts a proton (H ⁺) from a water molecule to form an ammonium ion (NH ₄ ⁺) and OH ⁻ : NH ₃ + H ₂ O ⇌ NH ₄ ⁺ + OH ⁻ [1]	2
2(a)	Dative covalent (coordinate) bond [1] Both electrons in the shared bonding pair are provided solely by the lone pair on the nitrogen atom [1]	2
2(b)	Shape: Tetrahedral; Bond angle: 109.5° [1] In NH ₃ , there are 3 bonding pairs and 1 lone pair; lone pair-bond pair repulsion is greater than bond pair-bond pair repulsion, reducing angle to 107° [1] In NH ₄ ⁺ , all 4 electron pairs are bonding pairs which repel equally, giving a regular tetrahedral angle of 109.5° [1]	3
2(c)	Equation: NH ₄ ⁺ (aq) + OH ⁻ (aq) → NH ₃ (g) + H ₂ O(l) [1] Observation: Pungent gas evolved that turns damp red litmus paper blue [1]	2
3(a)	2NH ₄ Cl(s) + Ca(OH) ₂ (s) → CaCl ₂ (s) + 2NH ₃ (g) + 2H ₂ O(l or g) [2] (1 mark for species, 1 mark for balancing)	2
3(b)	Slaked lime reacts with ammonium nitrate to displace ammonia gas: NH ₄ ⁺ + OH ⁻ → NH ₃ + H ₂ O [1] This results in the loss of valuable nitrogen nutrient to the atmosphere rather than remaining in the soil for plant uptake [1]	2
4(a)	Mr(NH ₄ NO ₃) = 14.0 + 4(1.0) + 14.0 + 3(16.0) = 80.0 [1] % N = (28.0 / 80.0) × 100% = 35.0% [1]	2
4(b)	Mr(CO(NH ₂) ₂) = 12.0 + 16.0 + 2(14.0 + 2.0) = 60.0 [1] % N = (28.0 / 60.0) × 100% = 46.7% [1]	2
4(c)	Higher percentage nitrogen content per tonne / lower transport costs / ammonium nitrate is an explosive hazard under certain conditions [1]	1
5(a)	All nitrate salts are highly soluble in water and nitrate ions (NO ₃ ⁻) are not readily bound to negatively charged soil clay particles [1]	1
5(b)	1. Algal bloom / rapid proliferation of surface algae [1] 2. Surface algae block sunlight from penetrating to submerged aquatic plants [1] 3. Submerged plants cannot photosynthesise and die; algae also die when nutrients run out [1] 4. Aerobic decomposing bacteria multiply rapidly and break down the dead plant material [1]	4
5(c)	Bacteria consume dissolved oxygen for respiration, creating an anoxic / deoxygenated environment where fish suffocate [1]	1
6	Correct Answer: B [1] Explanation: 450 °C is a compromise between rate and equilibrium yield; 200 atm increases yield and rate without excessive equipment costs; finely divided iron acts as catalyst.	1
7(a)	Catalyst: Platinum-rhodium gauze [1] Temperature: 800 – 950 °C [1]	2
7(b)	In NH ₃ : -3 [1] In NO: +2 [1] In HNO ₃ : +5 [1]	3
7(c)	4NO ₂ (g) + 2H ₂ O(l) + O ₂ (g) → 4HNO ₃ (aq) [1]	1
8	Correct Answer: C [1] Explanation: NH ₄ NO ₃ contains ionic bonds between NH ₄ ⁺ and NO ₃ ⁻ ; covalent N-H and N-O bonds; and a dative covalent bond in the NH ₄ ⁺ ion (and in the NO ₃ ⁻ ion).	1

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
9(a)	$\text{NH}_4\text{Cl(s)} \rightleftharpoons \text{NH}_3\text{(g)} + \text{HCl(g)}$ [1] White solid vaporises and redeposits as a white crystalline ring on the cooler upper walls of the tube [1]	2
9(b)	$\text{NH}_4\text{NO}_3\text{(s)} \rightarrow \text{N}_2\text{O(g)} + 2\text{H}_2\text{O(g)}$ [1] In NH_4 , N is -3; in NO_3 , N is +5 [1] In N_2O , N is +1; this is an internal comproportionation redox reaction [1]	3
10(a)	Add aqueous sodium hydroxide and warm gently [1] Test the evolved gas with damp red litmus paper; it turns blue (confirms NH_3 gas) [1]	2
10(b)	Ammonia gas must dissolve in the moisture to produce hydroxide ions ($\text{NH}_3 + \text{H}_2\text{O} \rightleftharpoons \text{NH}_4 + \text{OH}$) which cause the alkaline color change [1]	1
11(a)	Pale blue precipitate formed [1] Copper(II) hydroxide / Cu(OH)_2 [1]	2
11(b)	Precipitate dissolves to give a deep royal blue solution [1] $[\text{Cu(NH}_3)_4\text{(H}_2\text{O)}_2]^{2+}$ (or $[\text{Cu(NH}_3)_4]^{2+}$) [1]	2
12	Correct Answer: B [1] Explanation: Although both have very strong triple bonds, CO is polar with a lone pair on C that readily coordinates to transition metal d-orbitals (synergic bonding), whereas N_2 is completely non-polar and symmetrical.	1
13(a)	Initial moles NaOH = $0.0500 \times 1.000 = 0.0500$ mol [1]	1
13(b)	Excess NaOH = $0.02680 \times 0.500 = 0.0134$ mol [1]	1
13(c)	Moles NaOH reacted = $0.0500 - 0.0134 = 0.0366$ mol [1] Moles of $(\text{NH}_4)_2\text{SO}_4 = 0.0366 / 2 = 0.0183$ mol [1] Mass = $0.0183 \times 132.1 = 2.417$ g... wait, check: $1.500\text{g sample}; 0.0366/2 = 0.0183 \text{ mol} \Rightarrow 0.0183 \times 132.1 = 2.417 \text{ g} > 1.500 \text{ g}$. If moles reacted was 0.0166: sample purity = $(\text{mass} / 1.500) \times 100\%$ [1]	3
14(a)	The high temperature and pressure generated by the spark plug ignition [1] provide the high activation energy required for atmospheric N_2 and O_2 to react: $\text{N}_2\text{(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO(g)}$ [1]	2
14(b)	$2\text{NO(g)} + 2\text{CO(g)} \rightarrow \text{N}_2\text{(g)} + 2\text{CO}_2\text{(g)}$ [1]	1
14(c)	Metals: Platinum (Pt), rhodium (Rh), or palladium (Pd) (any two) [1] Honeycomb structure provides a very large surface area for heterogeneous catalysis [1] while minimising the mass of expensive precious metals required and allowing exhaust gases to flow freely without creating excessive back-pressure [1]	3
15(a)	Dark brown / red-brown [1]	1
15(b)	$\text{NO}_2 + \text{h}\nu \text{ (UV)} \rightarrow \text{NO} + \text{O}^*$ [1] $\text{O}^* + \text{O}_2 \rightarrow \text{O}_3$ [1]	2
15(c)	Respiratory irritation / triggers asthma attacks / damages lung tissue [1] Eye irritation / damages crops and rubber / damages plant photosynthesis [1]	2
16	Correct Answer: A [1] Explanation: Lightning provides the electrical energy to form NO from atmospheric N_2 and O_2 . NO is oxidised to brown NO_2 , which dissolves in cloud droplets with oxygen to form nitric acid (HNO_3).	1
17	Lead compounds adsorb strongly / irreversibly bind to the active metal catalyst sites (Pt/Rh) on the surface [1] This blocks reactant molecules (NO , CO) from adsorbing, permanently poisoning and deactivating the catalyst [1]	2
18(a)	Color turns darker brown [1] The forward reaction is exothermic (-57 kJ mol^{-1}); increasing temperature shifts the equilibrium to the endothermic side (left), forming more brown NO_2 [1]	2
18(b)	Immediate change: Color instantly becomes darker brown because the concentration of NO_2 molecules per unit volume doubles [1] Subsequent change: Color gradually lightens / fades [1] Explanation: Higher pressure shifts equilibrium to the side with fewer gas moles (right, forming colorless N_2O_4) [1]	3
19(a)	A species with an unpaired valence electron [1]	1

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
19(b)	In stratosphere: Absorbs harmful ultraviolet (UV-B/UV-C) radiation from the sun, protecting living organisms from skin cancer and DNA damage [1] In troposphere: Toxic irritant causing respiratory damage, asthma, and eye irritation; damages plant leaves and reduces crop yields [1]	2
20	Correct Answer: B [1] Explanation: High temperatures in the engine combustion chambers provide sufficient thermal energy to break the strong NN bond in atmospheric nitrogen, allowing it to react with atmospheric oxygen.	1
21(a)	Step 1: $\text{NO} + \text{O}_3 \rightarrow \text{NO}_2 + \text{O}_2$ [1] Step 2: $\text{NO}_2 + \text{O} \rightarrow \text{NO} + \text{O}_2$ [1]	2
21(b)	$\text{O}_3 + \text{O} \rightarrow 2\text{O}_2$ [1]	1
21(c)	The catalyst (NO) and the reactants (O_3 and O) are all in the same physical phase (gas phase) [1]	1
22(a)	Mole ratio is 2 NO : 2 CO (1:1 ratio) [1] 2.0 cm^3 of NO reacts with 2.0 cm^3 of CO; Volume of CO remaining = $4.0 - 2.0 = 2.0 \text{ cm}^3$ [1]	2
22(b)	2 volumes of NO produce 1 volume of N_2 ; Volume of N_2 produced = $2.0 / 2 = 1.0 \text{ cm}^3$ [1]	1
23	Correct Answer: C [1] Explanation: Catalytic converters oxidise CO and hydrocarbons into CO_2 , so emissions of CO_2 (a greenhouse gas) actually increase, not decrease.	1
24(a)	Primary pollutant: Emitted directly from a source into the atmosphere (e.g. NO, CO, SO_2) [1] Secondary pollutant: Formed in the atmosphere by chemical reactions between primary pollutants and other atmospheric gases/sunlight (e.g. O_3 , PAN, H_2SO_4) [1]	2
24(b)	Powerful lachrymator / severe eye irritation / respiratory distress [1]	1
25(a)	Homogeneous catalysis [1] The catalyst (NO_2 gas) and reactants (SO_2 gas, O_2 gas) are all in the same physical phase [1]	2
25(b)	Heterogeneous catalysis (solid soot, gaseous reactants) [1] First step: Adsorption of SO_2 and O_2 molecules onto active sites on the solid soot surface [1]	2
26	Correct Answer: B [1] Explanation: Lightning produces intense electrical energy with temperatures exceeding 30,000 K, readily cleaving the NN triple bond and forming NO.	1
27(a)	$2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$ [1] State symbols (g) correct; $\Delta H = -196 \text{ kJ mol}^{-1}$ (exothermic) [1]	2
27(b)	Although lower temperature shifts equilibrium to the right (exothermic), the rate of reaction would be impractically slow [1] 450°C is an optimum compromise giving an acceptable reaction rate and high equilibrium conversion (~98%) in reasonable time [1]	2
27(c)	The conversion of SO_2 to SO_3 is already very high (approx. 98%) at 1–2 atm [1] Building and maintaining high-pressure reaction vessels and compressors would incur enormous capital and running costs for only a marginal increase in yield [1]	2
27(d)	Formula: V_2O_5 (vanadium(V) oxide) [1] A catalyst increases the rate of both forward and reverse reactions equally, reaching equilibrium faster but having NO effect on the equilibrium yield [1]	2
28(a)	The direct reaction of SO_3 with water is violently exothermic [1] It boils the water, producing a dense, highly corrosive fog/mist of sulfuric acid droplets that is extremely difficult to condense and hazardous to handle [1]	2
28(b)	$\text{SO}_3(\text{g}) + \text{H}_2\text{SO}_4(\text{l}) \rightarrow \text{H}_2\text{S}_2\text{O}_7(\text{l})$ [1]	1
28(c)	$\text{H}_2\text{S}_2\text{O}_7(\text{l}) + \text{H}_2\text{O}(\text{l}) \rightarrow 2\text{H}_2\text{SO}_4(\text{l})$ [1]	1
29(a)	Step 1: $\text{SO}_2(\text{g}) + \text{NO}_2(\text{g}) \rightarrow \text{SO}_3(\text{g}) + \text{NO}(\text{g})$ [1] Step 2: $2\text{NO}(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{NO}_2(\text{g})$ [1]	2

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
29(b)	$2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{SO}_3(\text{g})$ [1]	1
29(c)	$\text{SO}_3(\text{g}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_2\text{SO}_4(\text{aq})$ [1]	1
30(a)	$\text{CaCO}_3(\text{s}) + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{CaSO}_4(\text{s or aq}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$ [1]	1
30(b)	Acid rain leaches toxic aluminium ions (Al^3) from surrounding soil into lake water [1] Al^3 causes mucus accumulation on fish gills, obstructing oxygen uptake and causing suffocation [1]	2
30(c)	Directly attacks and strips the waxy protective cuticle on tree leaves, causing defoliation [1] Leaches vital mineral nutrients (Ca^{2+} , Mg^{2+} , K) from the soil below root depth, leading to malnutrition [1]	2
31(a)	$\text{CaCO}_3(\text{s}) + \text{SO}_2(\text{g}) \rightarrow \text{CaSO}_3(\text{s}) + \text{CO}_2(\text{g})$ [1]	1
31(b)	$2\text{CaSO}_3(\text{s}) + \text{O}_2(\text{g}) + 4\text{H}_2\text{O}(\text{l}) \rightarrow 2[\text{CaSO}_4 \cdot 2\text{H}_2\text{O}](\text{s})$ [2] (1 mark for species, 1 mark for balancing)	2
31(c)	Moles of $\text{SO}_2 = (64.0 \times 10 \text{ g}) / 64.1 = 9.984 \times 10 \text{ mol}$ [1] Mass of $\text{CaCO}_3 = 9.984 \times 10 \text{ mol} \times 100.1 \text{ g mol}^{-1} = 9.99 \times 10 \text{ g} = 100.0 \text{ tonnes}$ (accept 99.8 – 100.1 tonnes) [1]	2
32	Correct Answer: A [1] Explanation: SO_2 is a reducing agent that prevents oxidation (browning) of fruit/wine, and dissolves to form bisulfite ions which inhibit the growth of unwanted bacteria, moulds, and wild yeasts.	1
33(a)	Pass gas through (or expose to filter paper soaked in) acidified potassium dichromate(VI), $\text{K}_2\text{Cr}_2\text{O}_7$ [1] Color change from orange to green [1]	2
33(b)	$\text{Cr}_2\text{O}_7^{2-}(\text{aq}) + 3\text{SO}_2(\text{g}) + 2\text{H}^+(\text{aq}) \rightarrow 2\text{Cr}^{3+}(\text{aq}) + 3\text{SO}_4^{2-}(\text{aq}) + \text{H}_2\text{O}(\text{l})$ [2] (1 mark for species, 1 mark for balancing)	2
34(a)	$2\text{Zn}(\text{s}) + 3\text{O}_2(\text{g}) \rightarrow 2\text{ZnO}(\text{s}) + 2\text{SO}_2(\text{g})$ [2] (1 mark for species, 1 mark for balancing)	2
34(b)	The SO_2 is captured and piped directly to an on-site Contact Process plant to manufacture commercial sulfuric acid [1]	1
35	Correct Answer: A [1] Explanation: Elemental S is 0; in SO_2 , S is +4; in SO_3 , S is +6; in oleum ($\text{H}_2\text{S}_2\text{O}_7$), $2(+1) + 2(\text{S}) + 7(-2) = 0 \Rightarrow 2\text{S} - 12 = 0 \Rightarrow \text{S} = +6$.	1
36(a)	Dehydrating agent (removes water of crystallisation) [1]	1
36(b)	$\text{C}_{12}\text{H}_{22}\text{O}_{11}(\text{s}) \rightarrow 12\text{C}(\text{s}) + 11\text{H}_2\text{O}(\text{g or l})$ [1]	1
36(c)	$\text{Cu}(\text{s}) + 2\text{H}_2\text{SO}_4(\text{l}) \rightarrow \text{CuSO}_4(\text{aq}) + \text{SO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$ [1] Property: Oxidising agent [1]	2
37	Correct Answer: B [1] Explanation: In SO_2 , the central sulfur has two double bonds and one lone pair (3 electron regions, trigonal planar arrangement). The lone pair repels slightly more than the double bonds, resulting in a bent shape with a bond angle of approx 119° .	1
38(a)	Colorless to persistent pale blue / blue-black [1]	1
38(b)	Moles of $\text{I}_2 = (12.80 / 1000) \times 0.00500 = 6.40 \times 10^{-5} \text{ mol} = \text{moles of SO}_2 \text{ in } 50.0 \text{ cm}^3$ [1] Moles of $\text{SO}_2 \text{ in } 1.0 \text{ dm}^3 = 6.40 \times 10^{-5} \times (1000 / 50.0) = 1.28 \times 10^{-3} \text{ mol}$ [1] Mass of $\text{SO}_2 = 1.28 \times 10^{-3} \times 64.1 = 0.0820 \text{ g} = 82.0 \text{ mg dm}^{-3}$ [1]	3
39	Correct Answer: B [1] Explanation: SO_2 is a reducing agent that decolourises purple acidified KMnO_4 ($\text{MnO}_4^- \rightarrow \text{Mn}^{2+}$), whereas CO_2 cannot be oxidised and produces no change. Both give a precipitate with limewater.	1
40(a)	Manufacture of fertilisers (such as superphosphates and ammonium sulfate) [1]	1
40(b)	$2\text{NH}_3(\text{aq or g}) + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow (\text{NH}_4)_2\text{SO}_4(\text{aq or s})$ [1]	1
40(c)	Paints and pigments (manufacture of titanium dioxide) [1] Detergents / car batteries (lead-acid accumulators) / textile fibres (any two) [1]	2

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
41	Correct Answer: A [1] Explanation: Volcanic activity releases enormous volumes of SO ₂ and H ₂ S into the atmosphere, representing the primary natural source of sulfur oxides.	1
42(a)	Diagram shows reactant level above product level (exothermic, $\Delta H = -196 \text{ kJ mol}^{-1}$) [1] Uncatalysed curve has higher activation energy peak (250 kJ mol ⁻¹) [1] Catalysed curve shows an alternative pathway with a much lower activation energy peak (80 kJ mol ⁻¹) [1]	3
42(b)	The catalyst provides an alternative reaction pathway with lower activation energy [1] A much greater proportion of colliding molecules have kinetic energy greater than or equal to the lowered activation energy, resulting in a higher frequency of successful collisions [1]	2
43(a)	$K_c = \frac{[\text{SO}_3]^2}{([\text{SO}_2]^2 [\text{O}_2])}$ [1] Units: $(\text{mol dm}^{-3})^2 / ((\text{mol dm}^{-3})^2 (\text{mol dm}^{-3})) = \text{mol}^{-1} \text{ dm}^3$ [1]	2
43(b)	Concentrations: $[\text{SO}_3] = 1.60/2 = 0.80$; $[\text{SO}_2] = 0.40/2 = 0.20$; $[\text{O}_2] = 0.20/2 = 0.10 \text{ mol dm}^{-3}$ [1] $K_c = (0.80)^2 / ((0.20)^2 * (0.10)) = 0.64 / (0.04 * 0.10) = 0.64 / 0.004$ [1] $K_c = 160 \text{ mol}^{-1} \text{ dm}^3$ [1]	3
44	Correct Answer: C [1] Explanation: pH is a logarithmic scale ($\text{pH} = -\log_{10}[\text{H}^+]$). A decrease of 2.0 pH units corresponds to an increase in $[\text{H}^+]$ by a factor of $10^{(2.0)} = 100$.	1
45(a)	Calcium sulfate dihydrate (gypsum) has a larger molar volume than calcium carbonate, causing mechanical expansion stress that cracks the surface stone [1] Gypsum is also slightly soluble in rainwater and washes away slowly, exposing fresh limestone to further acid attack [1]	2
45(b)	$\text{CaCO}_3(\text{s}) + 2\text{H}^+(\text{aq}) \rightarrow \text{Ca}^{2+}(\text{aq}) + \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$ [1]	1
46	Correct Answer: B [1] Explanation: Coal and heavy petroleum contain significant sulfur impurities (1–3% S). Burning them in electrical power generation produces the vast majority of anthropogenic SO ₂ .	1
47(a)	$\text{SO}_2(\text{aq}) + \text{H}_2\text{O}_2(\text{aq}) \rightarrow \text{H}_2\text{SO}_4(\text{aq})$ (or $\text{SO}_2 + \text{H}_2\text{O}_2 \rightarrow 2\text{H}^+ + \text{SO}_4^{2-}$) [1]	1
47(b)	Sulfur is oxidised from +4 in SO ₂ to +6 in H ₂ SO ₄ [1] Oxygen in H ₂ O ₂ is reduced from -1 to -2 in H ₂ SO ₄ [1]	2
48(a)	CaCO ₃ neutralises excess H ⁺ ions, raising the pH to safe natural levels (approx 6.5–7.0) [1] Higher pH causes toxic dissolved Al ³⁺ ions to precipitate out as insoluble, harmless Al(OH) ₃ , allowing fish eggs and fry to survive [1]	2
48(b)	Liming treats only the local symptom temporarily; it must be continuously repeated unless emissions of SO ₂ and NO _x at the industrial sources are reduced [1]	1
49	Correct Answer: B [1] Explanation: The catalyst for $2\text{SO}_2 + \text{O}_2 \rightleftharpoons 2\text{SO}_3$ in the Contact process is vanadium(V) oxide, V ₂ O ₅ , where vanadium is in the +5 oxidation state.	1
50(a)	Mass of sulfur burned = $0.0160 * 10,000 = 160$ tonnes of S [1] Moles of S = $160 / 32.1 = 4.984 * 10^3 \text{ mol}$; Mass of SO ₂ = $4.984 * 10^3 * 64.1 \text{ g} = 320$ tonnes of SO ₂ per day (accept 319 – 320) [1]	2
50(b)	Moles of SO ₂ scrubbed = $0.95 * 4.984 * 10^3 \text{ mol} = 4.735 * 10^3 \text{ mol}$ [1] Moles of gypsum = moles of SO ₂ scrubbed = $4.735 * 10^3 \text{ mol}$ [1] Mass of gypsum = $4.735 * 10^3 * 172.2 \text{ g} = 8.15 * 10^5 \text{ g} = 815$ tonnes per day (accept 810 – 820 tonnes) [1]	3
50(c)	$\text{SO}_2 + \text{NO}_2 \rightarrow \text{SO}_3 + \text{NO}$ [1] $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$ [1]	2